

Frank Yang Dininno

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University of California, San Diego — B.S. Materials Science (Expected Jun 2026) GPA: 3.79/4.00
University of California, Los Angeles — M.S. Electrical Engineering (Admitted Class of 2028)

Technical Skills

Software: Python, Java, Bash, Linux, Conda, Pandas, NumPy, serial communication, open-source FEM & FDTD simulation tools (FEMWELL, MEEP), Lumerical, Quantum Espresso, Biochemical tools (RFdiffusion, Addgene, ChimeraX), CAD software (Blender, Autodesk Inventor, Onshape, Klayton, OpenSCAD, Gmsh).

Cleanroom: Nano3 certified and trained.

Metrology: Ellipsometer, SEM, AFM, optical microscopy, stylus profiler, thin-film stress monitor.

Deposition: PECVD, spin coating. **Etching:** ICP-RIE, SLR (plasma etchers). **Photolithography:** MLA (Heidelberg), photoresist development.

Shop: Milling machine, resin 3D printing, FDM 3D printing, general machining and fabrication, laser system alignment.

Languages: Mandarin, English

Work Experience

Silicon Nitride Waveguide Fabrication and Characterization — *Coworker*. Role: cleanroom process experimentalist. Completed etch tests on waveguide-patterned chips using the Oxford Plasmalab RIE-ICP machine to optimize dielectric waveguide sidewall dihedral angle. Characterized etch quality using the Ziess Sigma SEM machine and programmed MATLAB scripts to characterize RMS sidewall etch roughness. Treated sidewall roughness as spatial phase noise "gratings" to calculate scattering losses. Part of a larger initiative to study and fabricate photonic ring resonator structures at the UCSD Fainman Lab.

Projects

Graphene Compression Research (Senior Project) — *Researcher*. Designed and manufactured a two-tonne compression device in the UCSD metal shop to study graphene composites for thermal interface materials; achieved improved thermal conductivity in compressed structures. Confirmed final microstructure using an SEM.

Elastomeric Graphene Thermal-Interface Materials Research — *Primary Researcher*. Working to develop a hybrid elastomeric, vertically-aligned graphene composite for computer chip heat dissipation and ease of installation. Purchased machinery and developed sophisticated chemical process for industrial scale-up.

Quantum Dot Positioning Systems for Telecom — *Primary Researcher*. Developing a CMOS compatible and scalable method for precision positioning of quantum dots within photonic devices for single photon emission. Tools implemented include PECVD, ICP-RIE, ALE, ALD, colloidal quantum dots (2nm-10nm), surface functionalization chemistry.

Nano-Innovator-Grant Proposal: Nanosecond Switching Langmuir Nanogates — *Sole Drafter*. Theorized and proposed a photochemical-based nanopore array (included documentation in website portfolio) that introduced chemical headgroups into a solvent with nanosecond optical switching speeds.

Node Form (Blender Add-on) — *Sole Developer*. Software extension for Blender enabling mathematical animation generation through node-based procedural design. <https://github.com/FyDininno/NodeForm>

Eagle Project — *Designer & Builder*. Constructed a mobile caged garden bed system (included documentation in website portfolio); managed design, materials, and build process.

Classes Taken

Semiconductor Devices; Cleanroom Device Fab Laboratory; Nanoelectronics; Semiconductor Device Materials Computation and Fabrication; Micro/Nanofabrication; Photolithography & Microfabrication; Nanomaterials Synthesis Lab; Nano Systems Characterization; Materials Science; Thermodynamics of Materials; Polymeric Materials; Crystallography; Solid State Physics; Density Functional Theory; Electronic Devices & Circuits; Systems and Signals; Linear Control Theory; Laser Electronics; Electromagnetic Optics; Optical Information Processing & Holography; Waves & Classical Optics; Heat and Mass Transport; Relativity & Quantum Physics; Heat Transport; Physical Chemistry; Biochemical Principles; Organic Chemistry; General Chemistry Lab; Nuclear Physics Principles; Differential Equations; Honors Calculus (31AH/BH/CH); Abstract Algebra; MATLAB for Engineers.

Honors & Leadership & Interests

Achieved the rank of **Eagle Scout** (Nov 2021, Troop 730). **FIRST Robotics CAD Lead** – 3256 Warriorborgs (2021 World's Advancement). Listed contributor of *Geometric Algebra for Electrical Engineers* by Peter Joot (<https://peeterjoot.com/archives/math2025/GAelectrodynamics.color.V0.3.6.4.pdf>). Likes backpacking, 500 miles trekked. Enjoys playing piano. Enjoys philosophy.